EcoSynapse

Miniature MRF Prototype for Indian Mixed Waste

Complete Technical Manual — Revision 3.1 (Final, Integrated)

Branched Dry/Wet Processing Cascade with Closed-Loop Item Tracking

Pinakin Sharma
B.Tech Electrical Engineering, IIT Bombay
Roll No. 25B1218, Batch 2029
IDEAS Level 1, Cohort 18, DSSE

July 3, 2026

Abstract

This document is the complete, self-contained build manual for the EcoSynapse bench-scale miniature Material Recovery Facility (MRF) prototype. It supersedes and merges Revision 2, Revision 3, and the Revision 3 addendum into a single Revision 3.1 reference. Revision 2 introduced mechanical pre-processing (shredding, dewatering, fine screening, air classification) to correctly handle wet Indian mixed waste. Revision 3 resolved a sequencing conflict in Revision 2 by branching the cascade immediately after hazard-reject into a dry/rigid path (intact-item sensor and vision sorting) and a wet/entangled path (shred and dewater only what needs it), and added a dedicated tramp-metal guard, a vibratory feeder for belt singulation, and a second moisture sensor at the dewatering press outlet. This Revision 3.1 manual folds in the previously separate addendum — coarse-trommel instrumentation (Section 4.1.6) and closed-loop encoder-based item tracking for gate actuation timing (Section 4.6) — and adds two new reference sections not present in any prior revision: a per-stage engineering deep-dive (Section 9) and a detailed CAD modelling roadmap (Section 10), so that a single document is sufficient to design, simulate, build, and validate the entire machine.

Contents

1 Why Revision 3

1.1 The continuity problem in Revision 2

Revision 2's cascade was a single straight line: hazard-reject → shred → dewater → fine-screen → air-classify → magnet → inductive → capacitive → camera → servo gate. This mixes two incompatible processing philosophies in one path:

- Stages 1–4 are built to *destroy* item identity — shredding opens bags and reduces every object, plastic bottle and metal can included, to fragments so that liquid can be pressed out and fines screened off.
- Stages 5–9 are built to act on *intact, individually distinguishable* items — a camera cannot classify "plastic vs. cardboard" from a jumbled stream of co-mingled shredded fragments, and a single servo paddle cannot cleanly route a continuous fragment stream into five separate bins.

Running recyclables through the destructive stages before asking the intact-item sorting stages to classify what remains is not a labelling issue — it is a structural contradiction.

1.2 The Revision 3 fix: branch the cascade

A passive size/rigidity split is introduced immediately after hazard-reject. Rigid, discrete items (bottles, cans, folded cardboard) never enter the shredder at all — they continue whole to a singulation and sensor-fusion sorting line. Only soft, wet, entangled mass (curry slurry, wet paper, tea powder, small food scraps) proceeds through shredding and dewatering, exiting to a dedicated wet-organic/reject bin rather than re-joining the dry sorting line.

Hopper (Stage 0, Hazard Reject)
↓
Stage 0.5: Coarse Split Trommel (passive, size/rigidity)
↙ ↘
Wet path: Stage 0.75 Tramp-Metal Guard → Stage 1 Shredder → . . .
Dry path: Vibratory Feeder (singulation) → Main Belt

Why the split can be passive (no extra actuator). A coarse trommel with large slots (40–60 mm) sized well above any individual recyclable's smallest dimension performs the split by geometry alone: rigid discrete items are too large/stiff to fall through and continue along the drum to the dry-path exit; soft, wet, deformable, or already-small material falls through the slots into the wet-path funnel below. This reuses the same trommel-screening principle already proven at the fine-screen stage, so no new actuation or sensing is required for the split itself (instrumentation for *logging* that split is added in Section 4.1.6).

2 Prototype Description

2.1 Design Brief

The prototype must demonstrate that Indian mixed waste requires two distinct treatments applied selectively, not one destructive pipeline applied uniformly: wet/entangled material needs mechanical dewatering and size reduction before disposal or composting, while dry/rigid recyclables need to be kept intact so that discrete-item sensing and vision classification can route them to the correct material-stream bin.

2.2 Physical Form Factor

The unit occupies a compact two-tier bench footprint, roughly 800 mm (L) × 380 mm (W) × 520 mm (H), keeping a single operator-visible footprint with one full transparent acrylic side panel.

- **Chassis:** 3D-printed corner brackets and joints with 6 mm acrylic/plywood panels.
- **Hopper with hazard-reject gate:** inductive sensor ring at the throat, solenoid-latched diverter flap to a labelled HAZARD bin, belt halt and audible alert on detection.
- **Coarse split trommel:** a short, wide-slot rotating drum (printed, ∼180 mm diameter, 220 mm long, 40–60 mm slot width) immediately downstream of the hopper, mounted at a slight incline. Rigid items continue along the drum to a dry-path chute at the lowered end; everything that passes through the slots drops into a funnel feeding the wet-path shredder inlet. Instrumented per Section 4.1.6.
- **Tramp-metal guard:** a second, independent inductive proximity sensor mounted at the throat of the wet-path funnel, immediately before the shredder inlet — not shared with, and not a substitute for, the Stage 0 hopper sensor.
- **Mini shredder stage:** dual counter-rotating printed rollers, blunt coarse teeth, PETG, guarded housing with interlock switch. Receives only the wet-path fraction.
- **Auger dewatering press:** with a capacitive moisture probe mounted at the cake outlet to log dewatering performance directly.
- **Fine trommel and air classifier:** process only the wet fraction, discharging to the WET-ORGANIC/REJECT bin.
- **Vibratory feeder:** a small printed tray on an eccentric-mass vibration motor, mounted between the dry-path chute and the main sorting belt, singulating rigid items so they arrive on the belt spaced apart rather than clustered.
- **Main sorting belt and sorting gate:** magnet, inductive, capacitive, ESP32-CAM/TinyML stages, servo-actuated paddle into METAL, PLASTIC, PAPER/CARDBOARD bins, now with closed-loop encoder-based timing (Section 4.6).
- **Control enclosure:** 170 mm × 110 mm × 55 mm, housing three motor driver channels and the I2C GPIO expander.

2.3 Functional Sorting Cascade

1. **Stage 0 — Hazard reject.** Sharp/metallic-tip objects diverted to HAZARD bin before anything else happens.

2. **Stage 0.5 — Coarse split.** Passive size/rigidity trommel separates rigid discrete items (dry path) from soft/wet/entangled mass (wet path). Instrumented for logging (Section 4.1.6).
3. **Stage 0.75 — Tramp-metal guard (wet path only).** Independent inductive sensor immediately before the shredder inlet halts the wet-path feed and diverts on any metal detection that slipped past the coarse split.
4. **Stage 1 — Shred (wet path only).** Detangles and reduces particle size of the wet fraction.
5. **Stage 2 — Dewater (wet path only).** Auger press removes free liquid; moisture sensor at the outlet logs dewatering efficiency.
6. **Stage 3 — Fine screen (wet path only).** Mini trommel removes fine particulates (tea powder, ash).
7. **Stage 4 — Air classify (wet path only).** Splits residual light/heavy fractions; both converge to WET-ORGANIC/REJECT.
8. **Vibratory singulation (dry path only).** Spaces rigid items onto the main belt one at a time.
9. **Stage 5 — Magnetic separation (dry path).** Ferrous metal lifted off the belt.
10. **Stage 6 — Inductive confirmation (dry path).** Remaining non-ferrous metal flagged.
11. **Stage 7 — Capacitive triage (dry path).** Residual surface moisture flagged (belt-level decision, distinct from the Stage 2 performance sensor).
12. **Stage 8 — AI vision classification (dry path).** ESP32-CAM + TinyML with TCS3200 cross-check.
13. **Stage 9 — Actuated sort (dry path).** Servo gate routes to METAL, PLASTIC, or PAPER/CARDBOARD, fired by encoder-tracked position (Section 4.6).

Medical/hazardous waste handling policy (unchanged). This prototype is a mechanical/electronic demonstrator, not a certified biohazard-handling device. Only blunt simulants (capped/dulled syringe bodies, empty medicine strips, taped-over sharp edges) should ever be used for testing or demonstration — never real used sharps or biological waste.

3 Bill of Materials (BOM)

3.1 Budget Summary

Category	Estimated Cost (Rs.)
Sensing & AI hardware	2,600 – 3,600
Actuation (6 motors + 2 servos + drivers)	2,500 – 3,400
Mechanical structure & 3D printing	2,600 – 3,500
Power & control electronics	1,730 – 2,300
Belt, bearings, gears, misc. mechanical	800 – 1,200
Wiring, connectors, fasteners, safety interlock hardware	800 – 1,100
Subtotal	**11,000 – 15,100**

Budget reality check. The itemised total (Section 3.2) comes to approximately Rs. 10,775 for the base Revision 3 design, rising to approximately **Rs. 11,065** once the closed-loop instrumentation additions from Section 4.1.6 and Section 4.6 (Rs. 290: encoder pair, trommel-exit beam-break, trommel moisture strip) are included — still comfortably inside the Rs. 11,000–15,100 budget band.

3.2 Detailed BOM

Item	Purpose / Notes	Qty	Unit (Rs.)	Subtotal (Rs.)
ESP32-CAM (AI-Thinker)	Vision classification, main coordination	1	750	750
FTDI/USB-TTL programmer	Flashing ESP32-CAM	1	150	150
Inductive proximity sensor (LJ12A3-4-Z/BX)	Hopper hazard-reject ring (Stage 0)	1	300	300
Inductive proximity sensor (2nd unit)	Main-belt non-ferrous confirmation (Stage 6)	1	300	300
Inductive proximity sensor (3rd unit)	Wet-path tramp-metal guard, immediately before shredder (Stage 0.75)	1	300	300
Capacitive soil-moisture sensor v1.2	Belt-level residual moisture triage (Stage 7)	1	90	90
Capacitive soil-moisture sensor v1.2 (2nd unit)	Auger outlet — dewatering performance logging (Stage 2)	1	90	90
TCS3200 colour sensor module	Plastic colour cross-check	1	380	380
Neodymium magnet strip/discs (N35)	Ferrous separation	2	80	160
Small squirrel-cage/blower fan (5–12V)	Air classification stage	1	200	200

Item	Purpose / Notes	Qty	Unit (Rs.)	Subtotal (Rs.)
SG90 micro servo	Sorting gate + hazard diverter flap	2	150	300
Geared DC motor, 12V high-torque	Shredder drive	1	550	550
Geared DC motor, low-rpm	Auger dewatering drive	1	350	350
Geared DC motor, low-rpm	Fine trommel drive	1	300	300
Geared DC motor, low-rpm	Coarse split trommel drive (Stage 0.5)	1	300	300
Eccentric-mass vibration motor, 12V	Vibratory singulation feeder	1	180	180
TT gear motor, 1:48	Main conveyor drive	1	90	90
L298N motor driver (dual-channel)	Drives paired motors (shredder+auger, fine+coarse trommel)	2	120	240
L298N motor driver (3rd unit)	Conveyor motor channel	1	120	120
N-channel MOSFET switch module	Unidirectional PWM drive for vibratory feeder	1	60	60
SSD1306 OLED display (0.96”, I2C)	Live stage status readout	1	250	250
Buzzer + status/hazard LEDs	Alerts, hazard-reject and tramp-metal events	1 set	80	80
Push buttons (calibration/reset/e-stop)	Manual controls	3	20	60
Lid/door magnetic safety interlock switch	Shredder guard interlock	1	100	100
I2C GPIO expander (PCF8574)	Resolves ESP32-CAM’s limited native GPIO count	1	150	150
12V/5A SMPS adapter	Main power, covers six motors’ combined worst-case draw	1	650	650
Buck converter (LM2596, 5V/3A)	Regulated 5V rail for logic/sensors	1	90	90
Breadboard + jumper wires	Prototyping	1 set	250	250
Perfboard/general PCB	Final control board	2	50	100
Rubber band belt / conveyor belting	Main conveyor surface	1	100	100
608ZZ bearings	Roller/drum/auger/coarse-trommel shaft support	10	25	250
Small pulleys/gears (printed + hobby gears)	Drive trains across all rotating stages	1 set	300	300

Item	Purpose / Notes	Qty	Unit (Rs.)	Subtotal (Rs.)
M3/M4 nuts, bolts, threaded inserts	Fastening across all stages	1 set	400	400
PLA filament	General printed parts, dry/wet chute geometry	1.1 kg	800/kg	880
PETG filament	Shredder rollers/teeth, auger, both trommels, vibratory tray	0.95 kg	900/kg	855
Acrylic/plywood sheet (6 mm)	Side panels, base plate	1	550	550
Small catch-tray/reservoir	Collects dewatered liquid and dry/wet split runoff	2	50	100
Wiring, heat-shrink, connectors, cable ties	General wiring/finishing	1 set	400	400
			Base Rev. 3 subtotal	**≈10,775**
IR break-beam pair (encoder)	Belt position reference, Stage 9 gate timing (Sec. 4.6)	1	100	100
IR break-beam pair (trommel exit)	Dry-path item count / size estimate (Sec. 4.1.6)	1	100	100
Capacitive strip (trommel floor)	Split-accuracy logging (Sec. 4.1.6)	1	90	90
			Total (Rev. 3.1, integrated)	**≈11,065**

3.3 Trim Options if Over Budget

- Drop the TCS3200 colour sensor (saves Rs. 380) — camera alone still functions, just with slightly lower plastic-colour disambiguation accuracy.
- Share the coarse split trommel and fine trommel on a single motor via a printed belt/gear linkage feeding both drums at different points along one shaft (saves Rs. 300, adds mechanical complexity and couples their speeds).
- Replace the vibratory feeder's dedicated vibration motor with a small offset-weight glued directly to the existing conveyor motor shaft extension (saves Rs. 240, removes independent speed tuning of the feeder).
- Substitute the tramp-metal guard sensor by reusing the Stage 6 main-belt inductive sensor's firmware logic at a second physical mount point — *not recommended*, since this reintroduces the exact risk this stage exists to remove; only consider if budget is a hard blocker.

4 Technical Design

4.1 Mechanical Design Notes

4.1.1 4.1.1 Coarse Split Trommel

A short, wide-slot drum (180 mm diameter, 220 mm long, 40–60 mm slot width) mounted at a slight incline directly beneath the hopper/hazard-reject outlet. Slot width is set well above the smallest dimension of any target recyclable so these items ride along the drum's interior surface to the lowered exit rather than falling through, while soft, wet, or already-small entangled material passes through the slots into the wet-path funnel beneath.

Known limitation. The split is a rigidity/size proxy, not a true material-type classifier — a large wet cardboard mass that has absorbed enough liquid to become limp may fall through with the wet fraction even though it is nominally a "dry" recyclable, and a small rigid item (a bottle cap) will fall through with the wet fraction regardless of moisture. This is acceptable: the downstream tramp-metal guard protects the shredder from any small rigid metal that ends up in the wet stream, and losing occasional small rigid items to the wet-organic bin is a minor yield loss, not a safety or correctness failure.

4.1.2 4.1.2 Tramp-Metal Guard

An inductive proximity sensor identical in type to the Stage 0 and Stage 6 units, mounted at the throat of the wet-path funnel immediately before the shredder rollers engage. On detection, firmware halts the wet-path feed motor(s) before the shredder motor is allowed to start, and raises the same hazard/alert pathway used by Stage 0. This sensor is deliberately independent of the Stage 0 hopper sensor and the coarse-split geometry, matching standard industrial practice of placing tramp-metal detectors immediately before crushers and shredders regardless of upstream screening.

4.1.3 4.1.3 Shredder, Auger Press, Fine Trommel, Air Classifier

Dual counter-rotating blunt-tooth PETG rollers with lid interlock; fixed-pitch printed Archimedes screw in a slotted tube; short slotted drum with 3–5 mm slots; small fan blowing perpendicular to a short free-fall path. The only addition over the original method is the capacitive moisture probe at the auger outlet.

4.1.4 4.1.4 Auger Outlet Moisture Sensor

A second capacitive soil-moisture-type probe mounted in light contact with the cake as it exits the auger, distinct in purpose from the Stage 7 belt-level sensor: this one logs a moisture-in/moisture-out performance figure for the press itself, while Stage 7 makes a per-item triage decision on the dry path.

4.1.5 4.1.5 Vibratory Feeder

A small printed tray, flexure-mounted (thin printed leaf springs or foam isolation pads) between the dry-path chute exit and the main sorting belt, driven by a 12V eccentric-mass vibration motor under PWM control via the MOSFET switch module. Vibration amplitude/frequency is tuned empirically so rigid items advance across the tray and drop onto the belt spaced apart rather than in clumps.

4.1.6 4.1.6 Coarse Trommel Instrumentation (Integrated from Addendum)

The coarse split trommel (Section 4.1.1) performs its dry/wet split entirely passively — geometry alone decides whether an item exits along the drum or falls through a slot. Without added sensing the firmware has no way of knowing how many items were split, in which direction, or whether the split behaved as expected. Two low-cost additions close this gap without adding any new actuation:

- **Dry-path exit beam-break.** A single IR LED / phototransistor pair mounted across the mouth of the drum's lowered (dry-path) exit. Every item that rides the drum to completion breaks this beam on its way out, giving the firmware an interrupt-driven count of dry-path items and, from beam-break duration at a known feed rate, a coarse size estimate. Approx. Rs. 100.
- **Drum-floor moisture pad.** A thin capacitive strip lining the lower interior of the drum flags when a rigid item exiting to the dry path was wet enough to register above a threshold. This does not redirect the item — there is no actuator at this stage to do so — but converts the "known limitation" in Section 4.1.1 from an assumed failure mode into a logged, quantifiable one.

Firmware integration. Both signals are non-time-critical and are routed through the PCF8574 I2C expander alongside the other non-interrupt lines (Section 4.3.1), rather than consuming a native GPIO. Beam-break count and moisture-flag rate are logged per test run and reported in the validation results, giving Finale judges a measured (not assumed) split-accuracy figure.

4.2 System Block Diagram

See the branched diagram in Section 1.2. The dry path additionally carries the encoder-based position-tracking signal introduced in Section 4.6, and the coarse trommel additionally carries the two logging signals from Section 4.1.6 — both routed through the PCF8574 expander except the encoder, which is interrupt-driven and native-GPIO (Section 4.6.3).

4.3 Electronics Architecture

4.3.1 4.3.1 GPIO Budget and the New Expander

The AI-Thinker ESP32-CAM exposes a small number of free GPIOs once the camera interface and PSRAM lines are reserved. Revision 3 adds a third inductive sensor, a second capacitive sensor (ADC), and a PWM line for the vibratory feeder — enough to exceed the native pin budget on its own, before the addendum's encoder and trommel-logging lines are even added.

Fix: I2C GPIO expander (PCF8574). Non-time-critical digital lines (status LEDs, buzzer, push buttons, one of the two inductive sensor digital outputs, and — per Section 4.1.6 — the trommel-exit beam-break and moisture-pad lines) are moved onto a PCF8574 I2C expander, sharing the same two I2C pins already used for the OLED display. This frees native GPIOs for the lines that genuinely need interrupt-capable or ADC-capable pins: the hazard-reject and tramp-metal inductive sensors, both capacitive moisture sensors, the vibratory feeder PWM line, and — per Section 4.6.3 — the belt encoder.

4.3.2 4.3.2 Compute and Motor Channels

- **L298N #1:** shredder motor + auger motor.
- **L298N #2:** fine trommel motor + coarse split trommel motor.
- **L298N #3:** conveyor motor (second channel spare for future expansion).
- **MOSFET switch module:** vibratory feeder motor (unidirectional PWM only).
- **ESP32-CAM native GPIO:** servo PWM (×2), hazard-reject inductive sensor (interrupt), tramp-metal inductive sensor (interrupt), both capacitive sensors (ADC), belt encoder (interrupt, Section 4.6.3), OLED + PCF8574 (I2C, shared bus), e-stop button (interrupt, highest priority).
- **PCF8574 expander:** main-belt inductive sensor digital line, TCS3200 digital lines, buzzer, hazard/status LEDs, calibration/reset push buttons, trommel-exit beam-break, trommel moisture-pad flag.

4.4 Power Budget

Subsystem	Voltage	Typical/Peak Curr
Shredder motor (stall condition)	12V	up to 1.5–2 A bri
Auger motor	12V	200–400 r
Fine trommel motor	12V	150–300 r
Coarse split trommel motor	12V	150–300 r
Vibratory feeder motor	12V	100–250 r
Conveyor TT motor	5–6V	150–250 r
Both servos (active)	5V	200–500 mA combi
ESP32-CAM (incl. Wi-Fi TX peaks)	5V	250–500 r
Sensors + OLED + PCF8574 + encoder + beam-breaks	5V	<140 r
Estimated system peak, 12V rail	≈3.2–3.8 A (shredder-stall worst case, all motors activ	

Power supply. A 12V/3A adapter has no adequate margin once the coarse trommel and vibratory feeder motors are included in the worst-case simultaneous load. A **12V/5A adapter** is specified to keep a safe margin above the ≈3.2–3.8 A worst case and avoid brown-out risk to the 5V logic rail during a shredder stall. The two additional IR beam-break pairs and the encoder add negligible current (a few mA each) and do not change this sizing.

4.5 Firmware Logic (High-Level)

1. Idle, all motors off, waiting for hopper-load trigger.
2. **Stage 0 check:** hopper inductive ring polled continuously; on hazard detection, latch diverter servo to HAZARD position, halt all downstream motors, alert, require manual reset.
3. **Stage 0.5:** coarse split trommel runs continuously at low speed whenever material is present; passive mechanical split, so firmware only keeps the drum turning while logging beam-break count and moisture-flag rate (Section 4.1.6).

4. **Stage 0.75 check (wet path only):** tramp-metal sensor polled continuously ahead of the shredder; on detection, halt the wet-path feed and shredder motor before engagement, alert, require manual clear.
5. **Stage 1–4 (wet path):** shredder run with current-sense stall-detect/reverse-retry logic; auger runs while shredded material is present, logging the outlet moisture sensor reading per batch; fine trommel and air classifier run continuously during active wet-path processing.
6. **Vibratory feeder (dry path):** runs continuously whenever material is present on the dry-path chute, tuned to a duty cycle producing visibly spaced single-item delivery onto the belt.
7. **8a. Encoder-tracked handoff (dry path, new):** on vibratory feeder exit-beam break, open a new item-tracking record; begin accumulating encoder pulses tagged to that record.
8. **8b. Sensor tagging (new):** each of Stage 5–8's readings is written to the currently open item-tracking record whose accumulated pulse count matches that stage's known belt position, not arrival order alone.
9. **Stage 9 (updated):** servo fires when the open record's accumulated pulse count reaches N_{fire} for the gate position, using the record's fused classification to select METAL / PLASTIC / PAPER-CARDBOARD; the record is closed and removed after firing.

4.6 Item Tracking & Gate Actuation Timing (Integrated from Addendum)

4.6.1 4.6.1 The gap this section closes

The servo-actuated paddle (Stage 9) routes each item to the correct bin once Stage 5–8 sensor fusion has produced a classification, but the camera (Stage 8) is mounted upstream of the gate; by the time a classification is available, the item has already travelled further down the belt. Firing the servo immediately on classification, or after a fixed guessed delay, both fail if belt speed varies (motor load, battery sag, friction) or if two items are spaced closer together than assumed.

4.6.2 4.6.2 Closed-loop position tracking

A slotted encoder disc is mounted on the conveyor's driven roller shaft, read by the same class of IR break-beam pair used at the trommel exit (Section 4.1.6), giving the firmware a pulse train proportional to actual belt travel rather than an assumed constant speed.

- Each pulse corresponds to a fixed belt-surface distance Δd (set by disc slot count and roller circumference).
- When an item crosses the vibratory feeder's exit (a known, fixed position on the belt, x_0), the firmware timestamps it and begins accumulating encoder pulses for that item specifically.
- As the item passes each sensor station (magnet, inductive, capacitive, camera), the corresponding stage's readings are tagged against the item's accumulated encoder count, not against wall-clock time — so classification data stays correctly associated with the physical item even if belt speed changes mid-run.
- The gate fires when accumulated pulses since x_0 equal the known encoder-count distance from x_0 to the gate position x_g:

$$N_{\text{fire}} = \frac{x_g - x_0}{\Delta d}$$

Why this matters for singulation. The vibratory feeder already spaces items apart in time; the encoder makes that spacing meaningful in *distance*, so the firmware can distinguish item A at position N from item B at position N even under belt-speed drift, rather than relying purely on the assumption that items never bunch up.

4.6.3 4.6.3 Hardware and GPIO impact

The encoder beam-break pair is interrupt-driven (it must not miss pulses at belt speed) and is added to the native GPIO set alongside the two tramp-metal/hazard inductive sensors — these are the only three lines in the whole design that must never be serviced through the polled PCF8574 bus, since all three represent either a safety stop or a position reference that cannot tolerate I2C polling latency.

Component	Cost	Purpose
IR break-beam pair (encoder)	Rs. 100	Belt position reference, Stage 9 gate timing
IR break-beam pair (trommel exit)	Rs. 100	Dry-path item count / size estimate
Capacitive strip (trommel floor)	Rs. 90	Split-accuracy logging
Total addition	**Rs. 290**	

4.6.4 4.6.4 Firmware logic update

See steps 8a, 8b, and the updated Stage 9 in Section 4.5 above.

4.6.5 4.6.5 Validation addition

The Wokwi simulation plan (Section 5.1) is extended to include a virtual encoder input (a simulated pulse train at a settable rate) alongside the existing virtual sensors, specifically to validate that: (a) two items entering closer together than the vibratory feeder's tuned spacing do not have their sensor readings cross-assigned between tracking records, and (b) gate timing remains correct under a simulated belt-speed step change mid-item-travel, which an open-loop timer-based design would fail by construction.

5 Simulation and Validation Plan

5.1 Wokwi (firmware, GPIO, and dual-path sequencing logic)

Validates the branched control logic explicitly: hopper trigger → hazard check → coarse-split (passive, with logging per Section 4.1.6) → parallel wet-path (tramp-metal check → shredder stall-retry → auger/trommel timing) and dry-path (vibratory feed rate → encoder-tracked sensor fusion → servo gate, per Section 4.6) branches running concurrently, using virtual buttons/potentiometers and a virtual pulse-train generator standing in for real sensors, the encoder, and motor current readings.

5.2 Tinkercad Circuits (breadboard-level validation)

Verifies the full breadboard: three inductive sensors, two capacitive sensors, TCS3200, three L298N-class driver channels, one MOSFET switch module, PCF8574 expander, two IR beam-break pairs, one encoder input, and the ESP32-CAM sharing I2C and native GPIO without conflicts — catching wiring/pin-budget problems here, especially around the expander's address and interrupt-vs-polled line split, is far cheaper than discovering them after soldering.

5.3 LTspice (power electronics under six-motor load)

- Simulate the 12V/5A supply under a shredder-stall current step with the coarse trommel and vibratory feeder also active, confirming the adapter and wiring gauge don't sag enough to brown-out the 5V buck converter.
- Simulate the LM2596 buck stage's transient response to a simultaneous ESP32-CAM Wi-Fi TX spike and servo actuation, the worst-case combined logic-rail event.
- Model flyback/inductive-kick behaviour for all six DC motors across three L298N drivers plus the MOSFET-switched vibration motor, confirming diode/snubber adequacy on each.

5.4 KiCad (schematic capture and control-board layout)

Full schematic covering ESP32-CAM, PCF8574 expander, three L298N modules, one MOSFET switch module, six motors, two servos, five sensors (three inductive, two capacitive), TCS3200, two IR beam-break pairs, one encoder, OLED, buzzer/LEDs, and both safety interlocks on one sheet.

5.5 TinyML Model Validation (Edge Impulse)

Since only the dry path reaches the camera, training images must be collected from whole, singulated items exiting the vibratory feeder — not from shredded material. Intact items are easier to photograph consistently and easier for the model to learn than post-shredding fragments would be.

6 Additional Components Identified Through Research

- **Vibratory feeder for singulation:** the standard infeed mechanism ahead of every commercial optical/vision sorting line — its role is specifically to space items one at a time onto a belt or chute so a downstream camera and actuator can act on individual items.
- **Dedicated tramp-metal detection immediately before a shredder:** standard industrial practice independent of any upstream screening — metal detectors are placed immediately

before crushers/shredders/grinders specifically because upstream separation (magnets, screens) is never assumed to be 100% reliable.

- **I2C GPIO expanders (PCF8574) for ESP32-CAM projects:** the standard fix for the AI-Thinker board's limited free-GPIO count once a project grows past a handful of sensors/actuators, avoiding a board swap.
- **PETG for the coarse split trommel and vibratory tray:** extending the same wet/high-stress material rationale already applied to the shredder rollers, auger screw, and fine trommel.
- **Slotted-disc optical encoders for low-cost belt position sensing:** the standard low-cost substitute for a commercial rotary encoder module in a bench prototype — a printed slotted disc read by a single IR break-beam pair gives adequate resolution for item-level position tracking at these belt speeds.

7 3D-Printed vs. Purchased Components

7.1 3D-Printed Components

- Hopper funnel with hazard-diverter flap mount
- Coarse split trommel drum, wide-slot, with drive-gear mount
- Wet-path funnel and tramp-metal sensor mount
- Dry-path chute
- Shredder rollers and blunt teeth (PETG)
- Shredder guard housing with interlock switch mount
- Auger screw and slotted outer tube (PETG)
- Auger catch-tray/reservoir
- Auger outlet moisture-probe mount
- Fine trommel drum, slotted, with drive-gear mount (PETG)
- Air-classifier chamber and fan mount, light/heavy lane dividers
- Vibratory feeder tray with flexure mounts
- Conveyor rollers (driven + idler) and motor bracket
- Magnet cantilever mount
- All three inductive sensor mounts (hopper, tramp-metal guard, belt)
- Capacitive sensor mounts (belt-level and auger-outlet, two units)
- ESP32-CAM enclosure with lens cutout and LED ring
- Camera gantry/overhead mount
- Sorting-gate servo bracket and paddle
- Multi-way chute assembly (5 bins including hazard)

- Five removable output bins/trays
- Control electronics enclosure (base + lid, OLED window)
- Chassis corner brackets and panel joints
- Cable channels/clips throughout
- Trommel-exit beam-break bracket (emitter + receiver arms)
- Trommel-floor moisture-pad recess/mount
- Conveyor-roller encoder disc and beam-break bracket

7.2 Items to Purchase

- ESP32-CAM module + FTDI programmer
- Three inductive proximity sensors
- Two capacitive soil-moisture sensors
- TCS3200 colour sensor module
- Neodymium magnets
- Small blower/squirrel-cage fan
- Two SG90 servos
- Shredder geared DC motor (12V, high-torque)
- Auger, fine-trommel and coarse-trommel geared DC motors (low-rpm)
- Eccentric-mass vibration motor
- TT gear motor (conveyor)
- Three L298N motor driver modules
- N-channel MOSFET switch module
- PCF8574 I2C GPIO expander
- SSD1306 OLED display
- Buzzer, LEDs, push buttons
- Magnetic safety interlock switch
- 12V/5A adapter + 5V buck converter
- Breadboard, jumper wires, perfboard
- Rubber band belt material
- Bearings (608ZZ $\times$ 10)
- Small hobby gears/pulleys for drive trains
- M3/M4 fasteners and threaded inserts

- Acrylic/plywood sheet
- PLA and PETG filament
- Wiring, heat-shrink, connectors, cable ties
- Two IR LED/phototransistor break-beam pairs (encoder + trommel exit)

8 Complete Build Roadmap

Phase	Task	Deliverable / Validation
Day 1–2	Finalise BOM, place all orders (motors, sensors, drivers, PCF8574, IR pairs first — longest lead items)	Orders confirmed; filament (PLA + PETG) in hand
Day 2–5	CAD design of all mechanical stages (see Section 10 for the detailed per-part CAD plan): coarse split trommel, wet-path funnel, dry-path chute, vibratory tray, encoder disc, plus base stages (shredder, auger, fine trommel, air classifier, hopper, chassis, bins)	Complete sliced STL set for every stage
Day 5–6	Simulation phase 1: Wokwi — full dual-path state machine including encoder pulse-train sim and trommel logging	Working Wokwi project simulating complete branched firmware logic
Day 6–7	Simulation phase 2: Tinkercad breadboard layout for all sensors, three L298N drivers, MOSFET module, PCF8574 expander, encoder, beam-breaks — GPIO pin-budget check	Verified wiring plan with no pin conflicts
Day 7–8	Simulation phase 3: LTspice — 12V/5A rail under six-motor worst-case load; 5V buck converter under combined Wi-Fi+servo load	Confirmed component sizing, no brown-out risk
Day 8–9	Simulation phase 4: KiCad — full schematic capture	Clean schematic, used as build reference
Day 6–12	3D print all mechanical parts (longest-running task, run in parallel with simulation work)	All printed parts complete
Day 12–13	Assemble chassis, hopper with diverter, coarse split trommel (with beam-break + moisture pad mounted), wet-path funnel with tramp-metal sensor mount	Mechanically complete Stage 0–0.75 assembly, both inductive sensors and trommel logging sensors mounted and verified
Day 13–14	Assemble shredder with guard interlock, auger press with outlet moisture-probe mount	Stage 1–2 mechanically complete
Day 14–15	Assemble fine trommel drum and air-classifier chamber; empirically tune fall height/fan power	Visible light/heavy fraction separation achieved, wet path mechanically complete
Day 15–16	Assemble vibratory feeder tray and tune amplitude/frequency for single-item spacing	Visible singulated item delivery onto belt

Phase	Task	Deliverable / Validation
Day 16–17	Assemble main conveyor belt, magnet mount, belt-level inductive sensor, belt-level capacitive sensor mount, encoder disc and beam-break bracket	Stage 5–7 mechanically complete, encoder producing a clean pulse train on a hand-turned roller
Day 17–18	Bench-test every motor and sensor individually — including encoder and both trommel-logging sensors — against the Wokwi-validated firmware logic on real ESP32-CAM hardware	Each subsystem confirmed working in isolation on real hardware
Day 18–19	Collect TinyML training images from whole, singulated items exiting the vibratory feeder; train and deploy Edge Impulse model	Deployed on-device classifier with acceptable validation accuracy
Day 19–20	Full system integration: wire everything to the perfboard per the KiCad schematic, including PCF8574 expander and encoder; mount enlarged control enclosure	Fully wired, fully assembled dual-path machine
Day 20–21	Implement and bench-verify the encoder-tracked item-record logic (Section 4.6.4, steps 8a/8b) with real belt runs before full end-to-end trials	N_{fire} calibrated against measured belt speed; gate fires on the correct item under varying speed
Day 21–23	End-to-end trials with real mixed test waste (simulated slurry, entangled bags, tea powder, metal, plastic, blunt hazard simulants, tramp-metal simulants in the wet stream) — tune every threshold and timing	Measured per-stage and per-class performance, including tramp-metal guard trigger rate and trommel split-accuracy log
Day 23–24	Safety pass: verify hazard-reject and tramp-metal guard both reliably trigger, verify shredder guard interlock stops motor instantly on opening	Signed-off safety checklist
Day 24–25	Finishing: labelling, cable management, OLED status polish, capture demo photos/video	Demo-ready finished unit
Day 25–26	Documentation update, rehearsal, buffer for debugging	Finalised report + rehearsed demo

Roadmap philosophy. The fully integrated Rev. 3.1 design adds roughly 3–4 days over the original Revision 2 22-day plan, concentrated in the new mechanical assembly (coarse split, tramp-metal mount, vibratory feeder, encoder), one additional integration day dedicated specifically to calibrating the encoder-based gate timing, and an expanded safety pass covering two independent metal-detection events. Simulation work remains front-loaded and runs in parallel with the 3D-printing queue.

9 Per-Stage Engineering Deep-Dive (New)

This section gives a single-reference technical breakdown of every stage: the physical principle at work, the exact sensing/actuation involved (if any), and the firmware handling. Use this as the primary reference when explaining "how does stage X actually work" to judges, teammates, or your own future self mid-build.

9.1 Stage 0 — Hopper & Hazard Reject

Principle: passive inductive sensing ring at the hopper throat, checking every item as it falls past on the way in. **Mechanism:** an LC coil generates an oscillating field; a sharp/metallic-tipped object disturbs it and the sensor outputs HIGH. **Action:** firmware latches a solenoid-driven diverter flap to route the item to the HAZARD bin *before* it can reach any moving part, halts all downstream motors, sounds the buzzer, and requires manual reset — this is a fail-safe stop, not a routing decision like Stage 9's.

9.2 Stage 0.5 — Coarse Split Trommel

Principle: pure geometry, no sensing decision. A rotating drum with 40–60 mm slots, inclined downhill. Rigid/large items cannot fit through a slot regardless of orientation and roll to the low, open end (dry path); small/soft/wet items fall through the slots into a funnel (wet path). **New instrumentation (Section 4.1.6):** an IR beam-break at the dry-path exit counts items and estimates size from beam-break duration; a capacitive strip on the drum floor flags dry-path items that were wet enough to be borderline misroutes. Neither signal changes the routing — there is no actuator here — they only make the passive split's performance measurable.

9.3 Stage 0.75 — Tramp-Metal Guard

Principle: identical inductive sensing to Stage 0/6, but positioned at the point of no return — immediately before the shredder rollers. **Rationale:** upstream screening (Stage 0, Stage 0.5) is never assumed 100% reliable, matching standard industrial practice for crushers/shredders. **Action:** on detection, firmware halts the wet-path feed motor and prevents the shredder motor from engaging, raising the same alert pathway as Stage 0.

9.4 Stage 1 — Shredder

Principle: dual counter-rotating blunt-tooth PETG rollers detangle and reduce the wet fraction's particle size, opening bags and breaking up entangled mass so the auger can act on it. **Safety:** guarded housing with a magnetic lid interlock switch — opening the lid cuts power to the motor immediately, independent of firmware. **Control:** current-sense stall-detection with a reverse-retry cycle if the motor draws above a threshold for longer than expected (jam recovery without operator intervention for minor jams).

9.5 Stage 2 — Auger Dewatering Press

Principle: a fixed-pitch printed Archimedes screw inside a slotted tube compresses the shredded wet mass as it advances, squeezing free liquid out through the slots while compacting the solid cake toward the outlet. **New instrumentation:** a capacitive probe at the cake outlet gives a moisture-out reading per batch, which combined with an assumed/measured moisture-in figure gives a dewatering-efficiency metric for the pitch deck.

9.6 Stage 3 — Fine Trommel

Principle: the same trommel-screening principle as Stage 0.5, scaled down — a short slotted drum (3–5 mm slots) removes fine particulates (tea powder, ash) from the dewatered cake before air classification.

9.7 Stage 4 — Air Classifier

Principle: a small fan blows perpendicular to a short free-fall path as the fine-screened material drops; light fractions (paper fines, light plastic film) are blown sideways into one lane while heavier fractions fall more vertically into another. Both converge to the WET-ORGANIC/REJECT bin — this stage's purpose is characterising the wet-fraction composition for the pitch data, not creating a third bin.

9.8 Vibratory Singulation Feeder

Principle: a printed tray on flexure mounts, driven by an eccentric-mass vibration motor under PWM. Tuned amplitude/frequency causes rigid items to advance across the tray via directional micro-hops (the standard principle behind every commercial vibratory bowl feeder) and drop onto the belt spaced apart. **Why it matters:** singulation in time is a prerequisite for the encoder-based tracking in Section 4.6 to work — if two items arrive on the belt touching, their sensor readings cannot be reliably separated into two tracking records regardless of encoder resolution.

9.9 Stage 5 — Magnetic Separation

Principle: purely passive — a fixed neodymium magnet strip mounted close to the belt surface physically pulls ferrous metal off as it passes beneath. **No native sensing**, though an optional Hall-effect sensor can be added near the magnet if logging a "ferrous item removed" event is wanted for the validation data.

9.10 Stage 6 — Inductive Confirmation

Principle: an LC-coil inductive sensor identical in type to Stage 0/0.75, positioned to catch non-ferrous metal (e.g. aluminium cans) that the Stage 5 magnet cannot grab. **Output:** a binary "metal present" flag, routed through the PCF8574 expander (non-interrupt-critical) and tagged to the item's encoder-tracked position per Section 4.6.2.

9.11 Stage 7 — Capacitive Triage

Principle: the belt-level capacitive probe measures dielectric change near the item — wet material registers a different capacitance than dry material. **Output:** an analog moisture reading (ADC), distinct from the Stage 2 performance sensor, used as one input to the Stage 9 fusion decision (e.g. flagging a borderline-wet dry-path item).

9.12 Stage 8 — AI Vision Classification

Principle: ESP32-CAM captures a snapshot as the item passes; an on-device TinyML model (trained per Section 5.5) predicts material class, cross-checked against the TCS3200 colour sensor's reading for plastic-colour disambiguation. **Output:** the primary material-class prediction, tagged to the item's tracking record.

9.13 Stage 9 — Actuated Sort

Principle (updated per Section 4.6): the servo-actuated paddle fires not on a fixed timer but when the item's tracking record's accumulated encoder pulse count reaches N_{fire}, using the fused classification from Stages 5–8 to select METAL / PLASTIC / PAPER-CARDBOARD. This closed-loop timing is the single largest accuracy lever in the whole dry path, since every upstream sensor's work is wasted if the paddle fires on the wrong item.

10 CAD Modelling Roadmap (New)

This section is the detailed CAD execution plan referenced by Day 2–5 of the Build Roadmap (Section 8). Model in Fusion 360 (consistent with prior TorqueX/EcoSynapse CAD work); export STLs at 0.2 mm layer-appropriate tolerances for PLA structural parts and 0.15 mm for PETG wear parts.

10.1 Modelling Order and Dependencies

Model in the order below — each stage's chute/mount geometry depends on the previous stage's exit geometry, so modelling out of order causes rework.

1. **Chassis shell first.** Lock the 800×380×520 mm envelope and the two-tier internal deck heights before any stage geometry, so every subsequent part's mounting bosses reference a fixed datum.
2. **Hopper + hazard diverter.** Model the throat opening to match the inductive sensor's rated sensing distance (check datasheet Sn before finalising throat width).
3. **Coarse split trommel + beam-break bracket + moisture-pad recess.** The drum's exit-mouth geometry must leave a clear, unobstructed line-of-sight slot for the IR pair (Section 4.1.6) — model the bracket as part of this assembly, not bolted on afterward, to avoid alignment drift.
4. **Wet-path funnel + tramp-metal sensor mount.** Funnel throat width should be sized to the tramp-metal sensor's sensing face, mounted flush per the datasheet's recommended standoff.
5. **Dry-path chute.** Exit geometry must match the vibratory tray's infeed edge height — model these two parts together or with shared reference geometry.
6. **Shredder rollers/teeth, guard housing + interlock mount.** Interlock switch placement must physically prevent lid closure from completing the circuit unless rollers are correctly meshed — verify with a motion study, not just static fit.
7. **Auger screw + tube + outlet moisture-probe mount.** Screw pitch and tube slot geometry carried over from the base design; only new part is the probe mount, positioned for light contact with the exiting cake without obstructing flow.
8. **Fine trommel + air classifier chamber.** Fan mount and fall-height are the two empirically-tuned dimensions — model with a parametric fall-height variable so it can be adjusted without full re-modelling during Day 14–15 tuning.
9. **Vibratory feeder tray + flexure mounts.** Flexure spring geometry (thin printed leaf springs) should be modelled as a separate parametric sub-part so stiffness can be iterated without reprinting the whole tray.
10. **Conveyor rollers, motor bracket, encoder disc + beam-break bracket.** The encoder disc's slot count is a CAD decision with a direct firmware consequence: more slots = finer position resolution but a higher pulse rate the ISR must keep up with. Model the disc parametrically (slot count as a variable) so this trade-off can be tuned after Day 17–18 bench testing shows actual achievable ISR response.
11. **Sensor mounts (magnet, inductive, capacitive ×2).** Standardise a common mounting-boss pattern across all sensor mounts so spares are interchangeable.

12. **ESP32-CAM enclosure + camera gantry.** Camera height and angle should be modelled against the belt width to guarantee full-item framing at the singulated spacing achieved in Day 15–16 tuning — this is worth a quick paper mockup check before committing to the printed gantry.
13. **Sorting gate servo bracket + paddle + multi-way chute + bins.** Paddle throw angle must be verified against all three lane widths in a motion study before printing, since a too-shallow throw can leave an item straddling two lanes.
14. **Control electronics enclosure.** Sized last, once the final component list (including PCF8574, 3×L298N, MOSFET module, encoder breakout) is confirmed, to avoid guessing internal volume.

10.2 File Organisation

- One Fusion 360 top-level assembly file referencing per-stage sub-assemblies (chassis, hopper-stage0, trommel-stage0.5, wetpath-stage0.75to4, dryside-vibratoryto9, control-enclosure).
- Parametric parameters exposed at the top-level assembly for the three empirically-tuned dimensions identified above (air-classifier fall height, vibratory flexure stiffness, encoder slot count) so Day 14–18 tuning does not require touching individual part files.
- STL export naming convention: `stageN_partname_rev.stl` (e.g. `stage0.5_trommel-drum_r1.stl`) to keep the sliced-part inventory traceable against the Section 8 roadmap's "complete sliced STL set" deliverable.

10.3 Tolerancing Notes

- Bearing bores (608ZZ, 22 mm OD): print at nominal + 0.1–0.15 mm to allow a light press fit without overheating the plastic on insertion.
- Threaded-insert bosses (M3/M4): size per standard heat-set insert boss tables, not raw thread diameter — undersized bosses are the most common cause of split bosses during insertion.
- Sensor mounting faces: hold flat within 0.2 mm across the sensing face footprint, since inductive/capacitive sensors are sensitivity-rated for a specific standoff distance from the target, and a warped mount changes effective sensing range.
- Trommel slot width: print test coupons at 40, 50, and 60 mm slot widths before committing to the full drum print, and pass real test items (a crushed can, a folded cardboard flap, a small bottle) through each to empirically confirm which width best separates your actual test-waste set — this is cheaper to get right in a coupon than in a full 220 mm drum reprint.